

Practical - 10

1. Inodes

```
harshika@siri:~$ gedit inode_test.txt
** (gedit:2021): WARNING **: 09:15:53.371: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:2021): WARNING **: 09:15:53.372: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
harshika@siri:~$ ls -li inode_test.txt
26089 inode_test.txt
harshika@siri:~$ stat inode_test.txt
  File: inode_test.txt
  size: 102          Blocks: 8          IO Block: 4096   regular file
Device: 8,48   Inode: 26089       Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/harshika)   Gid: ( 1000/harshika)
Access: 2026-10-01 09:16:31.517409746 +0000
Modify: 2026-10-01 09:16:31.549409740 +0000
Change: 2026-10-01 09:16:31.549409740 +0000
 Birth: 2026-10-01 09:16:31.517409746 +0000
harshika@siri:~$ find . -inum 26089
./inode_test.txt
harshika@siri:~$ gedit inode_test.txt
** (gedit:2228): WARNING **: 09:17:40.963: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:2228): WARNING **: 09:17:40.963: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
harshika@siri:~$
```

2. File Reading and Writing

```
harshika@siri:~$ gedit mmap_demo.c
** (gedit:2812): WARNING **: 09:24:33.459: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:2812): WARNING **: 09:24:33.461: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
harshika@siri:~$ gcc -Wall -g mmap_demo.c -o mmap_demo
harshika@siri:~$ ./mmap_demo
=====
PRACTICAL 10 - mmap() DEMO
=====
Enter source file name: source.txt
=====
1. TRADITIONAL read()/write()
=====
Traditional copy completed.
Time taken: 0.020343 seconds
=====
2. MEMORY-MAPPED mmap()
=====
mmap copy completed.
Time taken: 0.006792 seconds
=====
PERFORMANCE COMPARISON
=====
Traditional read/write : 0.020343 seconds
Memory-mapped mmap()   : 0.006792 seconds
=====
Files created:
traditional_copy.txt
mmap_copy.txt
Implementation comparison:
Traditional I/O:
- Uses read() and write()

1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <string.h>
4 #include <fcntl.h>
5 #include <unistd.h>
6 #include <sys/mman.h>
7 #include <sys/stat.h>
8 #include <time.h>
9
10 #define BUFFER_SIZE 4096
11
12
13 /*
14  GET TIME
15  ===== */
16
17 double get_time()
18 {
19     struct timespec ts;
20
21     clock_gettime(CLOCK_MONOTONIC, &ts);
22
23     return ts.tv_sec +
24            ts.tv_nsec / 1000000000.0;
25 }
26
27
28 /*
29  TRADITIONAL read()/write()
30  ===== */
31
32 int traditional_io(
33     const char *source,
34     const char *destination
35 )
36 {
37     int source_fd;
38     int destination_fd;
39
40     char buffer[BUFFER_SIZE];
41
42     ssize_t bytes_read;
43     ssize_t bytes_written;
44
45     source_fd = open(source, O_RDONLY);
46
47     if (source_fd == -1)
48     {
49         perror("open source");
50         return -1;
51     }
52
53     destination_fd = open(
54
```